

KHUSHI PRASHANT MEHTA

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PROFESSIONAL EXPERIENCE

ASML

May 2026 - Aug 2026

Machine Learning Intern

Los Angeles, CA

- Designed and deployed Pulse Agent, a full-stack GenAI application that lets business users ask questions in natural language, converts them to SQL on structured pulse data, and surfaces results in Power BI, replacing manual query and report requests.
- Partnered with business, logistics, and finance stakeholders to translate KPI and operational questions into Text-to-SQL requirements, supporting more effective natural-language access to business data
- Fine-tuned a large language model on curated pulse datasets and SQL procedures, improving domain-specific answer accuracy and traceability on business-critical queries.
- Architected a Retrieval-Augmented Generation (RAG) pipeline that consolidated unstructured process documentation from multiple teams into a single searchable knowledge base for the agent.
- Implemented SQL metadata tagging in the natural-language interface, increasing output transparency and query traceability.
- Built time-series forecasting on 24,300+ shipment records that surfaced a 28% Q4-over-Q1 transportation cost increase, and delivered an interactive Power BI dashboard (KPIs, cost per kg, lane/mode/carrier trends) to logistics and finance leadership.

USC Marshall School of Business

May 2025 - Present

Teaching Assistant

Los Angeles, CA

- Reviewed and debugged Python ML codebases for 40+ students, diagnosing logic defects, unhandled edge cases, and performance bottlenecks and guiding fixes through more efficient data structures and algorithmic approaches, improving code reliability and execution efficiency
- Supported development of end-to-end ML and NLP pipelines in Python using TensorFlow and scikit-learn, covering data preprocessing, feature engineering, model training, and evaluation on real-world datasets, enabling students to build and assess working machine learning solutions
- Mentored predictive-analytics projects from problem definition to working solution, helping students turn ambiguous business problems into well-scoped ML systems through sound model selection.
- Reinforced software engineering best practices (modular design, reusable code, systematic debugging) for production-quality ML code.

Hashtechy

Sep 2023 - Dec 2024

UI Front-End Designer

Ahmedabad, India

- Designed and built AI-driven front-end interfaces for the Currently social media platform, integrating ML model outputs into personalized UI components and contributing to growth to 500K+ downloads and a Shark Tank India feature.
- Wrote SQL queries and Python scripts to evaluate 100+ user A/B tests, using behavioral metrics to improve retention.
- Collaborated with ML engineers to turn model-training insights into front-end features and built data visualizations for executives, making model findings easier to understand and apply

Space Application Center, ISRO

Jan 2023 - Jul 2023

AI/ML Research Intern

Ahmedabad, India

- Benchmarked k-means and Random Forest against deep learning models for land-cover classification using ArcGIS, ArcPy, and factorization on multi-source satellite imagery, providing a basis for comparing model performance
- Designed and trained a U-Net CNN for semantic segmentation of glacier-covered land, using data augmentation, hyperparameter tuning, and fine-tuning to improve classification accuracy by 20-30%.
- Evaluated the U-Net model on 1,000+ Sentinel-2 images using IoU, Dice score, and accuracy with Python, GDAL, Rasterio, and NumPy, quantifying model performance across the satellite image dataset

EDUCATION

University of Southern California

Jan 2025 - Dec 2026

M.S., Computer Science

Los Angeles, CA

- Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Database Systems, Analysis of Algorithms, Artificial Intelligence

Indus University

Jul 2019 - Apr 2023

B.Tech, Information Technology

Ahmedabad, India

- Coursework:** Web Technology, Cloud Computing, Artificial Intelligence, Data Science

TECHNICAL SKILLS

- Languages:** Python, SQL, C/C++, JavaScript, HTML/CSS
- Libraries and Frameworks:** NumPy, Pandas, Matplotlib, Plotly, PySpark, Kafka, TensorFlow, Keras, PyTorch, Scikit-Learn, OpenCV, CUDA-, Hugging Face Transformers, OpenAI, Claude, Gemini, Ollama, LangChain, LangGraph, LlamaIndex, CrewAI, Tavily, SerpAPI, FAISS-, ChromaDB, FastAPI, Flask, Django, Streamlit, Gradio, REST API, Power BI, Tableau, PostgreSQL, MySQL, MongoDB, Docker, MLflow, PyTest, Git, GitHub, Bash, AWS S3, AWS Glue, AWS Bedrock, AWS Aurora, AWS DynamoDB, AWS MSK, Jupyter, ArcGIS, ArcPy, Figma
- Cloud and OS:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Azure, Windows, Linux, Unix
- Technical Skills:** Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Computer Vision, Large Language Models (LLM), LLM Fine-Tuning, Agentic AI, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Prompt Engineering, Time Series Forecasting, Statistics, Regression, A/B Testing, Mathematics, Data Science, Data Analytics, Data Visualization, Data Engineering, ETL Pipelines, Data Validation, Synthetic Data Generation, Recommender Systems, System Design, Data Structures, Database Management, CI/CD, Cloud Computing

PROJECTS

MCP-Powered AI Assistant

- Python, Model Context Protocol, Claude API, Groq, pytest
- Built an AI assistant on Anthropic's Model Context Protocol with a decoupled tool server (web search, file I/O, REST API) and LLM client, enabling Claude/Groq provider swaps with no code changes; secured tool execution via sandboxed file access.

Loan Approval Prediction

- Python, Scikit-learn, Streamlit, OpenAI API
- Trained a logistic regression model (81% accuracy, 600+ applications) with Streamlit and OpenAI explanations, cutting decision time 40%.

AI-Powered Travel Planning Assistant

- Python, Flask, OpenRouter, Mixtral
- Built a full-stack GenAI assistant that generates personalized itineraries from natural-language queries, with multi-turn memory.